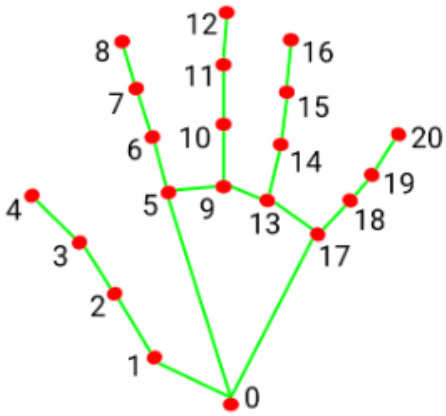
Hand detection

Hand detection
- 1. Hand detection
- 2. Effect preview
  - 2.1. USB camera

MediaPipe is a cross-platform machine learning framework that can process key point detection such as hands, faces, and postures in videos and images in real time.

1. Hand detection

MediaPipe hand detection can identify and track 21 key points of the hand.



0. WRIST
1. THUMB_CMC
2. THUMB_MCP
3. THUMB_IP
4. THUMB_TIP
5. INDEX_FINGER_MCP
6. INDEX_FINGER_PIP
7. INDEX_FINGER_DIP
8. INDEX_FINGER_TIP
9. MIDDLE_FINGER_MCP
10. MIDDLE_FINGER_PIP
11. MIDDLE_FINGER_DIP
12. MIDDLE_FINGER_TIP
13. RING_FINGER_MCP
14. RING_FINGER_PIP
15. RING_FINGER_DIP
16. RING_FINGER_TIP
17. PINKY_MCP
18. PINKY_PIP
19. PINKY_DIP
20. PINKY_TIP

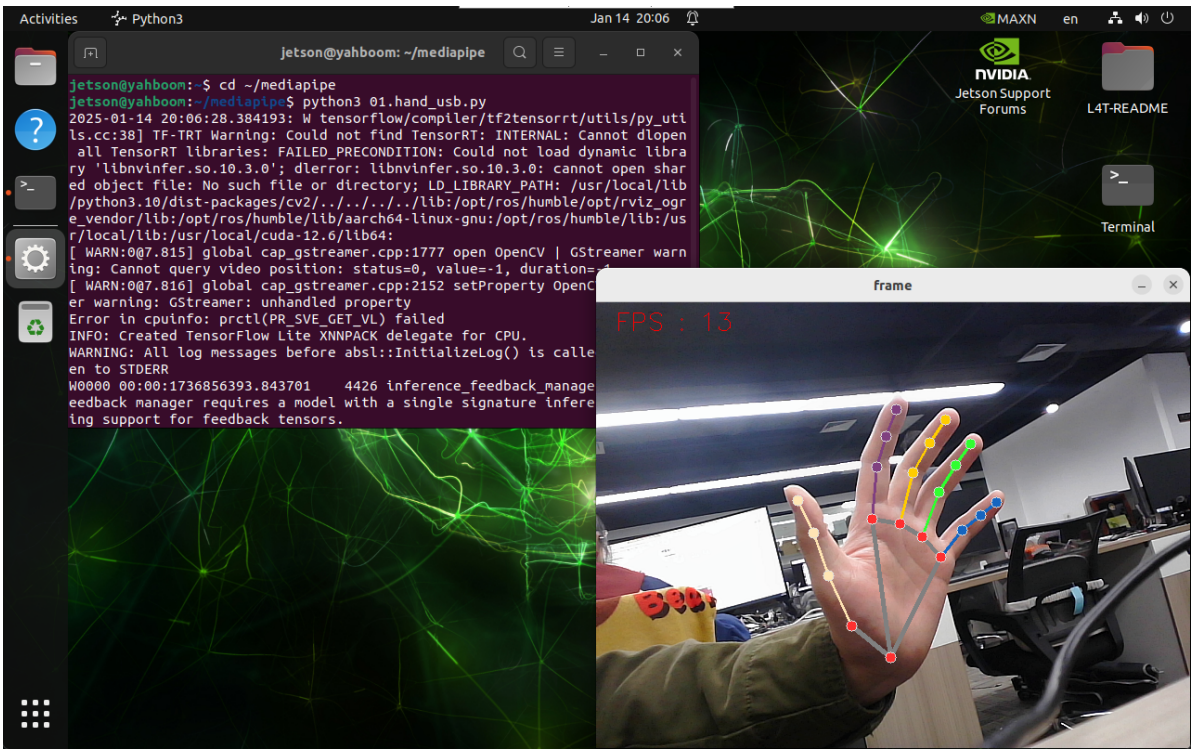
2. Effect preview

2.1. USB camera

```
cd ~/mediapipe
```

```
python3 01.hand_usb.py
```

Note: Select the camera image and press `q` to exit the program!


Activities
Python3
Jan 14 20:06
MAXN
en
jetson@yahboom: ~/mediapipe
jetson@yahboom:~$ cd ~/mediapipe
jetson@yahboom:~/mediapipe$ python3 01.hand_usb.py
2025-01-14 20:06:28.384193: W tensorflow/compiler/tf2tensorrt/utils/py_utils.cc:38] TF-TRT Warning: Could not find TensorRT: INTERNAL: Cannot dlopen all TensorRT libraries: FAILED_PRECONDITION: Could not load dynamic library 'libnvinfer.so.10.3.0'; dlerror: libnvinfer.so.10.3.0: cannot open shared object file: No such file or directory; LD_LIBRARY_PATH: /usr/local/lib/python3.10/dist-packages/cv2/../../../../lib:/opt/ros/humble/opt/rviz_ogre_vendor/lib:/opt/ros/humble/lib/aarch64-linux-gnu:/opt/ros/humble/lib:/usr/local/lib:/usr/local/cuda-12.6/lib64:
[WARN:0@7.815] global cap_gstreamer.cpp:1777 open OpenCV | GStreamer warning: Cannot query video position: status=0, value=-1, duration=
[WARN:0@7.816] global cap_gstreamer.cpp:2152 setProperty OpenCV warning: GStreamer: unhandled property
Error in cpuinfo: prctl(PR_SVE_GET_VL) failed
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
W0000 00:00:1736856393.843701 4426 inference_feedback_manager
eedback manager requires a model with a single signature inferencing support for feedback tensors.
NVIDIA
Jetson Support Forums
L4T-README
Terminal
frame
FPS : 13